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Systematic review and meta-analysis of single-stage versus two-stage revision for periprosthetic joint infection after knee arthroplasty: a call for a randomised trial

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- **Purpose:** Knee arthroplasty is an effective treatment for severe knee degeneration; however, periprosthetic joint infection (PJI) is one of its serious complications. Single- and two-stage revision are common treatments, but few studies have compared single- and two-stage revision for PJI after knee arthroplasty. This study aimed to compare the reinfection and reoperation rates of single- and two-stage revision through meta-analysis.
- **Methods:** The review process was conducted according to the PRISMA guidelines. We searched the PubMed, Medline, Embase and Cochrane Central Register of Controlled Trials databases for trials comparing single- and two-stage revision for PJI after knee arthroplasty from the respective inception dates to April 2023. Two researchers individually screened the studies, performed the literature quality evaluation and data extraction and used Stata 17 software for data analysis.
- **Results:** The meta-analysis showed that the reinfection rate was significantly lower in the single-stage revision group than in the two-stage revision group. While the reoperation rates demonstrated no statistically significant difference between the two groups. We presented descriptive results because the discrepancies in the knee function scores and data reported in the studies meant that these data could not be combined in the meta-analysis.
- **Conclusion:** Based on the available research, single-stage revision is a reliable option for PJI after knee arthroplasty. However, when developing the best treatment strategy, it is still necessary to consider the individual circumstances and needs of the patient, as well as the risks of postoperative rehabilitation and complications.

Keywords: periprosthetic joint infection; knee arthroplasty; single-stage revision; two-stage revision

Introduction

Knee arthroplasty is widely used to treat severe knee degeneration and is considered an effective measure that improves the quality of life and restores the mobility of patients (1, 2). However, despite the excellent

performance of knee arthroplasty in most patients, postoperative periprosthetic joint infection (PJI) (an infection that occurs in joint replacement) remains concerning as a rare but serious complication (3).

PJI is identified if: there is a sinus tract linked to the prosthesis, a pathogen is isolated from two separate joint samples, or four out of six specific criteria are met, including elevated blood markers, increased joint fluid cell counts, presence of joint purulence or positive culture or histologic findings from joint tissue (4). Infection can lead to prosthesis loosening, bone defects and even life-threatening conditions in severe cases (5). With advances in surgical techniques and improvements in surgical conditions, PJI currently occurs in less than 1% of patients with knee arthroplasty (6).

For patients with PJI after knee arthroplasty, it is critical to choose the most appropriate treatment strategy. Two-stage revision surgery was previously considered the gold standard for PJI (2); however, the ideal time interval between surgeries, choice of antimicrobial agent and timing of treatment remain controversial. Furthermore, the recurrence rate of PJI varies widely, and some patients are medically ill and may not tolerate a second surgery. In recent years, the single-stage revision technique has received widespread attention worldwide, and its application is increasing (7, 8). The shortcomings of two-stage revision can be avoided by performing single-stage revision, which is more conducive to the functional recovery of the affected limb, reduces the occurrence of complications, overall cost and surgical trauma, and improves the level of patient satisfaction (9). Blom *et al.* highlighted that single-stage revision showed cost-effectiveness over an 18-month follow-up period and is increasingly being adopted (10). Additionally, some centres report that single-stage revision has a success rate comparable to two-stage revision (11, 12, 13). However, there is a lack of clear evidence to identify the superior surgical modality between single- and two-stage revision for PJI after knee arthroplasty. Although some small clinical studies have explored this issue, their conclusions are inconsistent and limited by small sample sizes. Therefore, this review aimed to systematically pool and synthesise existing clinical research data by conducting a comprehensive meta-analysis to compare the efficacy of single- and two-stage revision for PJI after knee arthroplasty. Through this comprehensive analysis, we hope to provide more accurate and effective treatment recommendations for medical teams to optimise treatment strategies for patients with PJI after knee replacement, and contribute to the recovery and health improvement of patients.

Materials and methods

According to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement, this meta-analysis was performed in agreement. (14) The protocol for this meta-analysis was registered on PROSPERO (Registration No: CRD 42022384299).

Inclusion criteria

(i) Study type: randomised controlled trial, cohort study or case-control study. (ii) Study population: patients with infected knee arthroplasty. (iii) Intervention and control: single-stage in the treatment group and two-stage in the control group. (iv) Outcome index: reinfection rate, reoperation rate and knee function.

Exclusion criteria

(i) Letters, meetings, case reports, reviews or republished studies; (ii) studies lacking a control group; (iii) patients with septic arthritis or tuberculous arthritis; (iv) patients did not undergo second-stage in the two-stage group.

Outcomes

The primary outcome was the incidence of reinfection. A secondary outcome was the incidence of reoperation and knee function.

Search strategy

Two of the authors performed the search in PubMed, Medline, EMBASE and the Cochrane Central Register of Controlled Trials from the inception dates to April 30, 2023, using the keywords '(periprosthetic joint infection or prosthetic joint infection) and (Two-stage or 2-stage or two stage or second-stage or double-stage) and (Single-stage or one-stage or 1-stage) and knee and (unhealed or infection or reoperate* or revise or function or knee score or range of motion)'. No language restrictions were applied during the search.

Study selection

Two researchers screened the retrieved literature strictly and individually against inclusion and exclusion criteria. In cases where two researchers do not agree during the literature screening process, it will be left to the senior researcher.

Data collection process

Data on relevant outcome measures were extracted from the literature that met the inclusion criteria, including author year, country, age, study design, follow-up and outcomes by two researchers individually. If two researchers do not agree during the data collection, it will be left to the senior researcher.

Assessment of risk of bias and quality of evidence

Two researchers independently assessed the quality of all included trials based on Cochrane risk-of-bias

criteria (15). The Newcastle–Ottawa scale (NOS) was used to evaluate the literature quality of the retrospective studies (16). We also examined the quality of evidence for outcomes using the grading of recommendations assessment, development, and evaluation (GRADE) approach (17).

Data synthesis

The meta-analysis was performed using Stata (version 17; StataCorp, 2021) software. Heterogeneity was assessed by using the Q test and I^2 value calculation. If heterogeneity was not present ($P > 0.1$ and $I^2 < 50\%$), the data were combined using a fixed-effect model. The random-effects model was used if heterogeneity was present ($P < 0.1$ or $I^2 > 50\%$). In the investigation of potential publication bias, the Harbord test was employed to assess the presence of a small-study effect. Odds ratio (OR) (method of Mantel–Haenszel heterogeneity) and their associated 95% CIs were used to assess outcomes, with a P -value less than 0.05 indicating a statistically significant difference.

Sensitivity analyses

We performed a sensitivity analysis on a case-by-case exclusion basis.

Results

After identifying total of 279 studies, 142 duplicates were removed, and 99 irrelevant studies were excluded based on the titles and abstracts. The full text of 38 articles was read, and 27 retrospective studies were included in the systematic review (8, 11, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42). Due to six studies discussing hip and knee replacements (38, 38, 39, 40, 41, 42), we were unable to separate and extract knee-specific data for inclusion in our article. Consequently, we only included them in the systematic review, resulting in a final meta-analysis comprising 21 studies (8, 11, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36). The literature screening process is shown in Fig. 1. All included studies had NOS scores of greater than or equal to 6 points, indicating that they are high-quality studies. The basic characteristics of the included studies are shown in Table 1.

Reinfection rate

The 18 trials included in this study reported the reinfection rates after the two surgical regimens (11, 19, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36). As there was no heterogeneity between the results

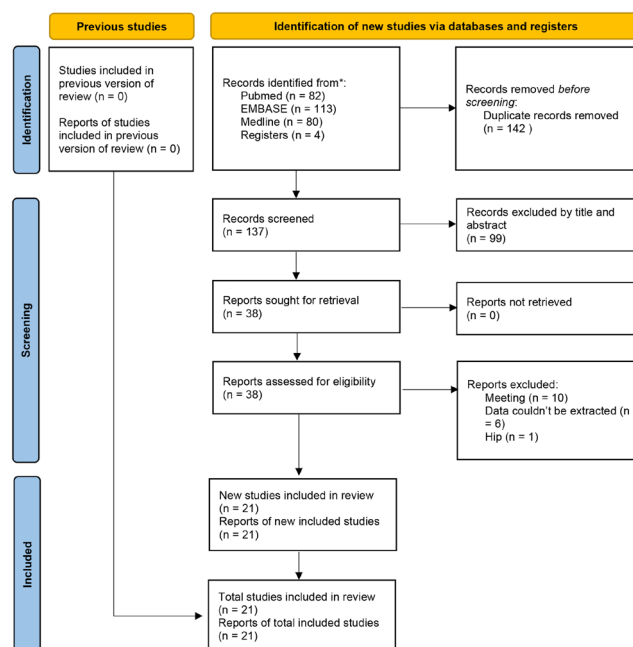


Figure 1

Flow diagram for search and selection of included studies.

($I^2 = 0\%$), the fixed-effect model was used to pool the results. The reinfection rate was significantly lower after single-stage revision than two-stage revision (OR = 0.72, 95% CI: 0.55–0.96, $P = 0.023$, Fig. 2).

Reoperation rate

A total of 10 studies reported the reoperation rates (8, 18, 21, 24, 26, 28, 30, 34, 35, 36). As there was no heterogeneity between study results ($I^2 = 0\%$), the fixed-effect model was used to pool the results. There was no statistically significant difference in the rate of reoperation after single-stage revision compared with two-stage revision (OR = 0.95, 95% CI: 0.77–1.17, $P = 0.240$, Fig. 3).

Knee function

Seven studies reported the postoperative knee function (18, 19, 20, 24, 30, 31, 34). The findings regarding the effects of single- and two-stage revision on knee function scores were heterogeneous and inconsistent and could not be combined in a meta-analysis. Therefore, we have presented the results descriptively and provided a brief summary of the study findings. Baker *et al.* found that the Oxford Knee Score did not significantly differ between the two groups (18). Bauer *et al.* also reported no significant difference between the two groups in the knee score but did not report the s.d. and CIs (19). Capuano *et al.* found no difference in Knee Society Scores (KSS) between the two groups at the final follow-up (20). Haddad *et al.* showed that the

Table 1 Characteristics of included studies.

Study	Country	Date range	Design	FU	Age*		Subjects, n		Outcomes	NOS score
					One stage	Two stage	One stage	Two stage		
Baker <i>et al.</i> (18)	UK	2008–2010	RCS	6–12 m	69.4 ± 10.7	70.3 ± 8.9	33	89	Reoperation, knee score, satisfaction, complications	8
Bauer <i>et al.</i> (19)	France	NA	RCS	2 Y	71.8	68.3	30	77	Reinfection rate, knee score	8
Capuano <i>et al.</i> (20)	Italy	2013–2015	RCS	29.3 M†	-	-	17	17	Reinfection, length of hospital stay, KSS	9
Chalmers <i>et al.</i> (21)	USA	2009–2017	RCS	1–9 Y	51–87	NA	1	4	Reinfection, reoperation	9
Crego <i>et al.</i> (22)	Germany	NA	RCS	NA			20	45	Reinfection	9
Cury <i>et al.</i> (23)	Brazil	2008–2010	RCS	NA			6	7	Reinfection, functionality score, life scores	6
Haddad <i>et al.</i> (24)	UK	2004–2009	RCS	3–9 Y	63 (48–87)	68 (45–85)	28	74	Reinfection, reoperation, knee score,	7
Kheir <i>et al.</i> (25)	USA	1991–2014	RCS	12 M	64.9 ± 9.2	65.4 ± 8.6	2	36	Failure rate	9
Klemm <i>et al.</i> (26)	USA	2015–2018	RCS	NA			44	88	Reinfection, reoperation, readmission, PROM	8
Laffer <i>et al.</i> (27)	Switzerland	1988–2003	RCS	2–193 M	70.1 (43.5–90.1)	69 ± 9	2	13	Success rate	8
Lenguerrand <i>et al.</i> (8)	Multicentre	2003–2014	RCS	NA			489	2377	Reoperation	9
Leta <i>et al.</i> (28)	Norway	1994–2016	RCS	1 Y			72	243	Reinfection, reoperation, survival rate, mortality rate	9
Li <i>et al.</i> (29)	China	2003–2014	RCS	12–158 M	71 (63–76)	67 (59–73)	22	87	Reinfection, complication	9
Massin <i>et al.</i> (30)	France	2005–2010	RCS	2 Y			108	177	Reinfection, reoperation, knee score	8
Matar <i>et al.</i> (11)	UK	2003–2018	RCS	2.0–17.6 Y	71.8 ± 9.8	70.5 ± 10.2	82	161	Success rate, survivorship rate	8
Ribes <i>et al.</i> (31)	France	2009–2014	RCS	1 Y	72.6 ± 9.2	69.5 ± 9.1	21	41	Reinfection, knee score	9
Ritter and Farris (32)	Indiana	1969–2004	RCS	1 Y			3	51	Success rate	9
Shanmugasundaram <i>et al.</i> (33)	Multicentre	2008–2011	RCS	NA			5	16	Success rate	6
Siddiqi <i>et al.</i> (34)	USA	2012–2017	RCS	2 Y	NA	NA	57	137	Reinfection, reimplantation, reoperation rates, success rate, ROM	9
Tuecking <i>et al.</i> (35)	Germany	2013–2019	RCS	18–92 M	65.0 ± 10.2	69.3 ± 11.1	15	48	Reinfection, reoperation, implant survival	9
Van den Kieboom <i>et al.</i> (36)	USA	2010–2018	RCS	2.5–22.9 Y	67.9 ± 10.6	65.0 ± 11.0	18	48	Reinfection, reoperation, amputation, readmission, mortality, hospital stay	9

*Data are mean ± s.d. or range; †Mean value.
NOS, Newcastle–Ottawa scale; FU, follow-up; M, month; Y, year; NA, not applicable; RCS, retrospective cohort study; KSS, knee society scores; PROM, patient-reported outcome measures; ROM, range of motion.

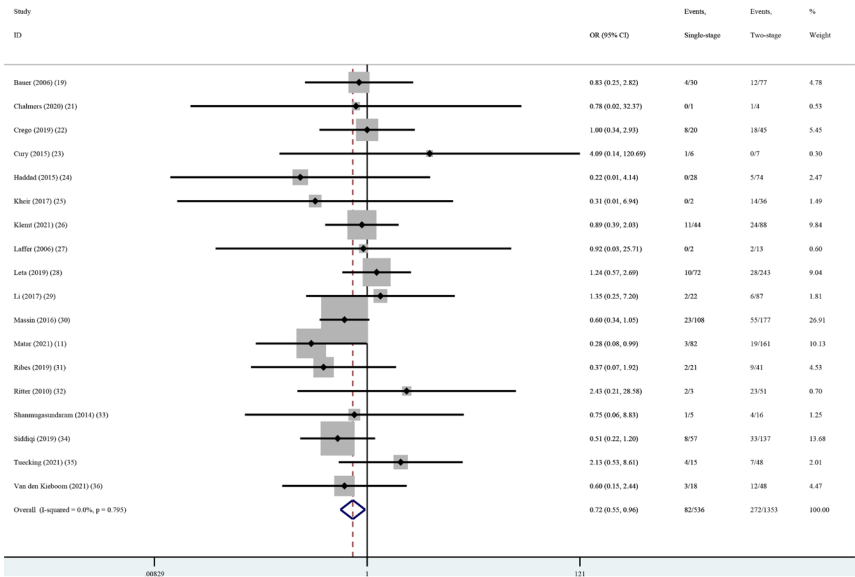


Figure 2
Reinfection rate in included studies.

knee function was better after single-stage revision than after two-stage revision based on the KSS but only showed the mean and range (24). Massin *et al.* reported that international KSS were similar between the two groups (30). Ribes *et al.* found that the international KSS and Knee Injury and Osteoarthritis Outcome Score were better in the single-stage revision group than the two-stage revision group (31). Siddiqi *et al.* showed that the postoperative knee range of motion did not significantly differ between the two groups (34). The detail of knee joint function scores for each study are shown in Table 2.

Sensitivity analysis

The remaining studies were combined using the OR values when any individual study was excluded. No individual study had a significant impact on the results (Fig. 4).

Risk of bias

As shown in Fig. 5, the funnel plots showed some asymmetry, which may suggest the presence of publication bias. However, the Harbord test showed no evidence of a small study effect regarding reinfection ($P = 0.66$) and reoperation ($P = 0.24$).

Discussion

Our meta-analysis results suggest that the reinfection rate is significantly lower after single-stage revision than after two-stage revision for PJI after knee arthroplasty. Two-stage revision has long been considered the gold standard for PJI, but it significantly reduces the activity time of patients, predisposing them to bone degeneration and periprosthetic fractures (43). Our

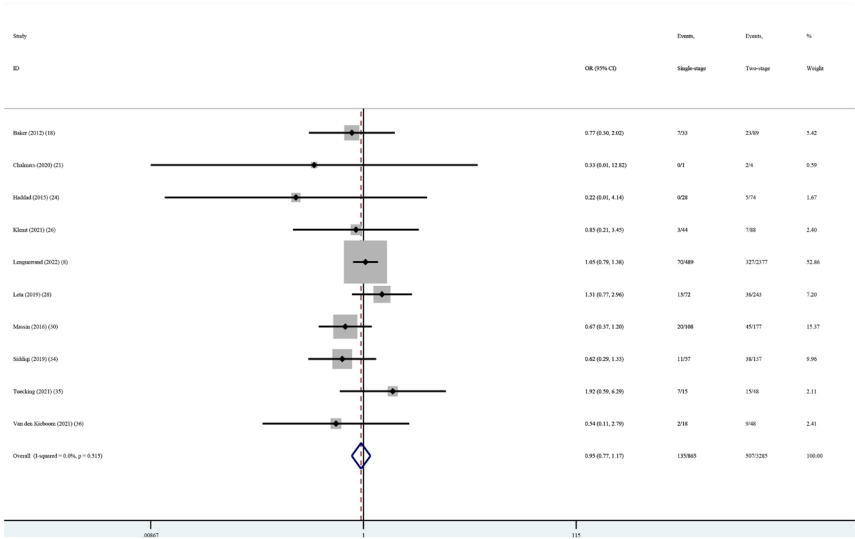


Figure 3
Reoperation rate in included studies.

Table 2 The results of the studies about knee function. Data are presented mean ± s.d. or as mean (range).

	Study						
	(18)	(19)	(20)	(24)	(30)	(31)	(34)
OKS							
One stage	24.9 ± 13.1						
Two stage	22.8 ± 12.4						
KSS							
One stage		75.5	78 ± 6.1	88 (38–97)			
Two stage		74.8	77.3 ± 6.4	76 (29–93)			
IKS							
One stage					88.6 ± 9.4	51.05	
Two stage					89.7 ± 2	39.75	
KOOS							
One stage						10	
Two stage						8.75	
SF-12							
One stage						39.1	
Two stage						34.41	
Flexion range							
One stage		92.5 (10–125)			97 ± 18		106.6 ± 23.2
Two stage		93 (20–125)			91 ± 24		104.0 ± 18.1

OKS, Oxford Knee Score; IKS, International Knee Society; KSS, Knee Society Score; KOOS, Knee Injury and Osteoarthritis Outcome Score; SF-12, Short form-12 questionnaire.

results suggest that single-stage revision is no less effective than two-stage revision and has a lower rate of infection recurrence. However, the studies included in our review were retrospective studies with some bias in the process of patient allocation, as the decision whether to perform single- or two-stage revision is made at the discretion of the surgeon after considering all the details of the patient and the surgical site. The need for larger-scale, randomised clinical trials to validate our conclusions and to comprehensively address potential biases in future investigations. Besides, single-stage revision can avoid the shortcomings of two-stage revision, reduce the hospitalisation time and cost, reduce the number of surgeries and improve patient satisfaction and quality of life (44, 45). Although recurrent infection seriously decreases quality of life, not all recurrent infections require reoperation; mild infections can be improved with anti-infective treatment. Our analysis found no difference in the reoperation rate after the two treatment modalities. Due to differences in reporting across studies, we were unable to pool the

knee function data; therefore, we could only describe the results. Although descriptive results are still useful to understand trends in knee function scores in the single- and two-revision groups, future studies should increase the sample sizes and add data from more independent studies to fully assess the knee function of patients with PJI after knee arthroplasty.

Previous meta-analyses of single- and two-stage revision for PJI after knee arthroplasty have been reported, but most have been single-arm meta-analyses that systematically reviewed the recurrence of infection after both surgical modalities (46, 47, 48). In contrast, we included studies that compared single- versus two-stage revision, providing more specific and reliable estimates of the outcomes of single- and two-stage revision for PJI after knee arthroplasty. Nagra *et al.* also compared the efficacy of single- and two-stage revision in the treatment of PJI in 2015 (49), but their study included only five cohort studies of 231 patients. Based on the included studies, their conclusion was

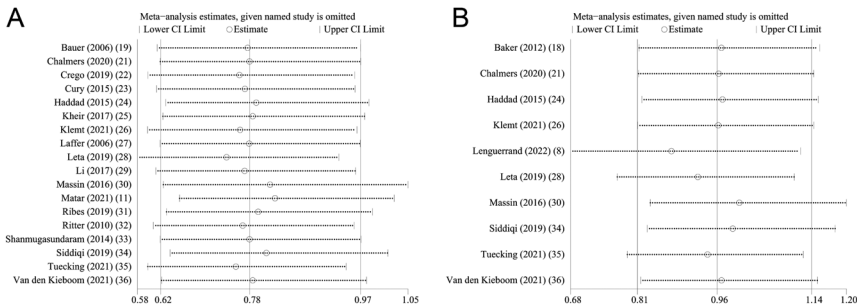
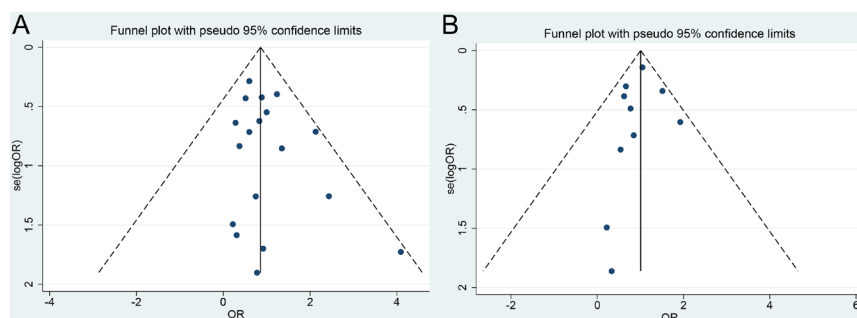


Figure 4
The result of sensitivity analysis of reinfection (A) and reoperation (B).

**Figure 5**

Funnel plot of the included studies in this meta-analysis for the incidence of reinfection (A) and reoperation (B).

that there was no difference in the re-infection rates between single-stage and two-stage revisions. However, their subgroup analysis revealed a divergence: among studies published before 2000 (including two studies), no difference was observed between single-stage and two-stage revisions. Contrastingly, among studies published after 2000 (comprising three studies), the re-infection rate was significantly lower in single-stage revisions compared to two-stage revisions. Since then, more studies comparing single- and two-stage revision have been published. As our results are based on a larger number of studies (21 studies, 4911 patients), our conclusion indicates that the re-infection rate in single-stage revisions is significantly lower than in two-stage revisions. Our study possesses stronger statistical power, robustly demonstrating these findings. Technological advancements, accumulated expertise and further optimisation of surgical techniques might contribute to the increased success rates in single-stage revisions. Additionally, we conducted a comparative analysis of reoperation rates between the two procedures, which was not performed in the study by Nagra *et al.*

Strengths and limitations

The present study has some strengths. First, we included a large number of studies (four times more than were included in the previous meta-analysis (49)) and compared the effectiveness of single- and two-stage revision in more detail. Secondly, in contrast to previous reviews (46, 47, 48), we used a two-arm meta-analysis approach that enabled the use of quantitative estimates to compare the effectiveness of the two revision strategies. Thirdly, our results remained robust in several sensitivity analyses.

Our study also had certain limitations. First, there was bias in the selection of the revision strategy. Capuano *et al.* used implants coated with an antibiotic-loaded hydrogel for single-stage revision but used uncoated implants for two-stage revision (20). Li *et al.* performed single-stage revision in cases with negative cultures and performed two-stage revision in cases with positive cultures (29). Secondly, the included studies were retrospective, and there was a lack of randomised controlled trials with large samples, resulting in reduced

confidence in the study results. Thirdly, when discussing the advantages and disadvantages of single- and two-stage revisions, we only considered reinfection rates, reoperation rates and knee function scores, and did not perform a comprehensive analysis of other relevant indicators such as patient satisfaction, quality of life and complication rates. Future studies should consider more factors to fully assess the advantages and disadvantages of the two surgical modalities.

Conclusion

The existing research suggests that single-stage revision surgery is a reliable option for PJI after knee arthroplasty. In clinical practice, medical professionals should carefully select the appropriate surgical method based on factors such as the individual characteristics of the patient, severity of infection and available surgical technology. We hope that our results will provide useful guidance for clinical decision-making and contribute to further improving the treatment outcomes and quality of life of patients with PJI after knee arthroplasty. Through continuous in-depth research, we have the potential to provide patients with safer and more effective treatment options, thereby promoting their recovery and wellbeing.

ICMJE Conflict of Interest Statement

The authors declare that there is no conflict of interest that could be perceived as prejudicing the impartiality of the study reported.

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Author contribution statement

CX and HL collected data, performed data analysis, interpreted results, and wrote the first draft of the manuscript. WP and SW reviewed the protocol, screened articles, extracted data, and reviewed the results and manuscript. XY and HL contributed to the systematic review protocol and critically reviewed the results and manuscript. HL contributed to the protocol development and reviewed the manuscript. All authors contributed to the conception and design of the study and reviewed all documents and materials.

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